Mini project report on

Operating System

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

UE24CS242B - Operating Systems

Submitted by:

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under the guidance of

Prof. Neha Sharma

Assistant Professor

PES University

Jan- May 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

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(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

CERTIFICATE

This is to certify that the mini project entitled

Operating Systems

is a bonafide work carried out by

<DEVOPAM PAL>	**<PES2UG24CS152>**
<Name2>	**<SRN2>**

In partial fulfilment for the completion of forth semester OS Project (UE24CS242B) in the Program of Study - Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Jan 2026 – May 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 4th semester academic requirements in respect of project work.

Signature

Prof. Neha Sharma

Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled **Operating System** has been carried out by us under the guidance of **Prof. Neha Sharma, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Jan – May 2026.

<name>	**<srn>**	**<Signature>**
<name>	**<srn>**	**<Signature>**

ACKNOWLEDGEMENT

I would like to express my gratitude to Prof. Neha Sharma, Department of Computer Science and Engineering, PES University, for her continuous guidance, assistance, and encouragement throughout the development of this UE24CS242B - Operating Systems Project.

I take this opportunity to thank Dr. Sandesh B J, C, Professor, Chairperson, Department of Computer Science and Engineering, PES University, for all the knowledge and support I have received from the department.

I am deeply grateful to Prof. Jawahar Doreswamy, Chancellor – PES University, Dr. Suryaprasad J, Vice-Chancellor, PES University for providing to me various opportunities and enlightenment every step of the way. Finally, this DBMS Project could not have been completed without the continual support and encouragement I have received from my family and friends.

ABSTRACT

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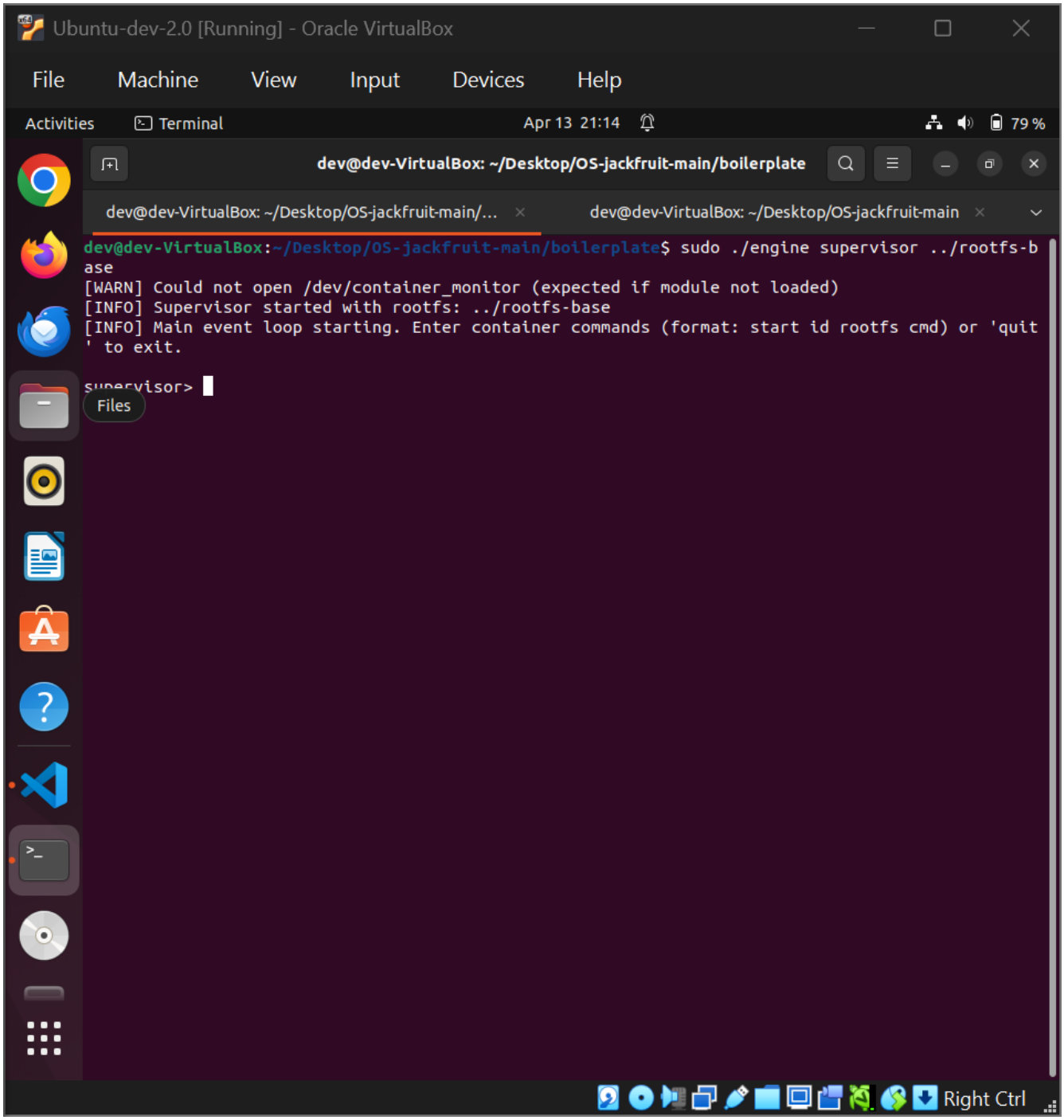
Phase 2: Core Runtime Architecture (User-Space engine.c)

TERMINAL:1



TERMINAL:2

Ubuntu-dev-2.0 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Activities Terminal Apr 13 21:14 79 %

dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main

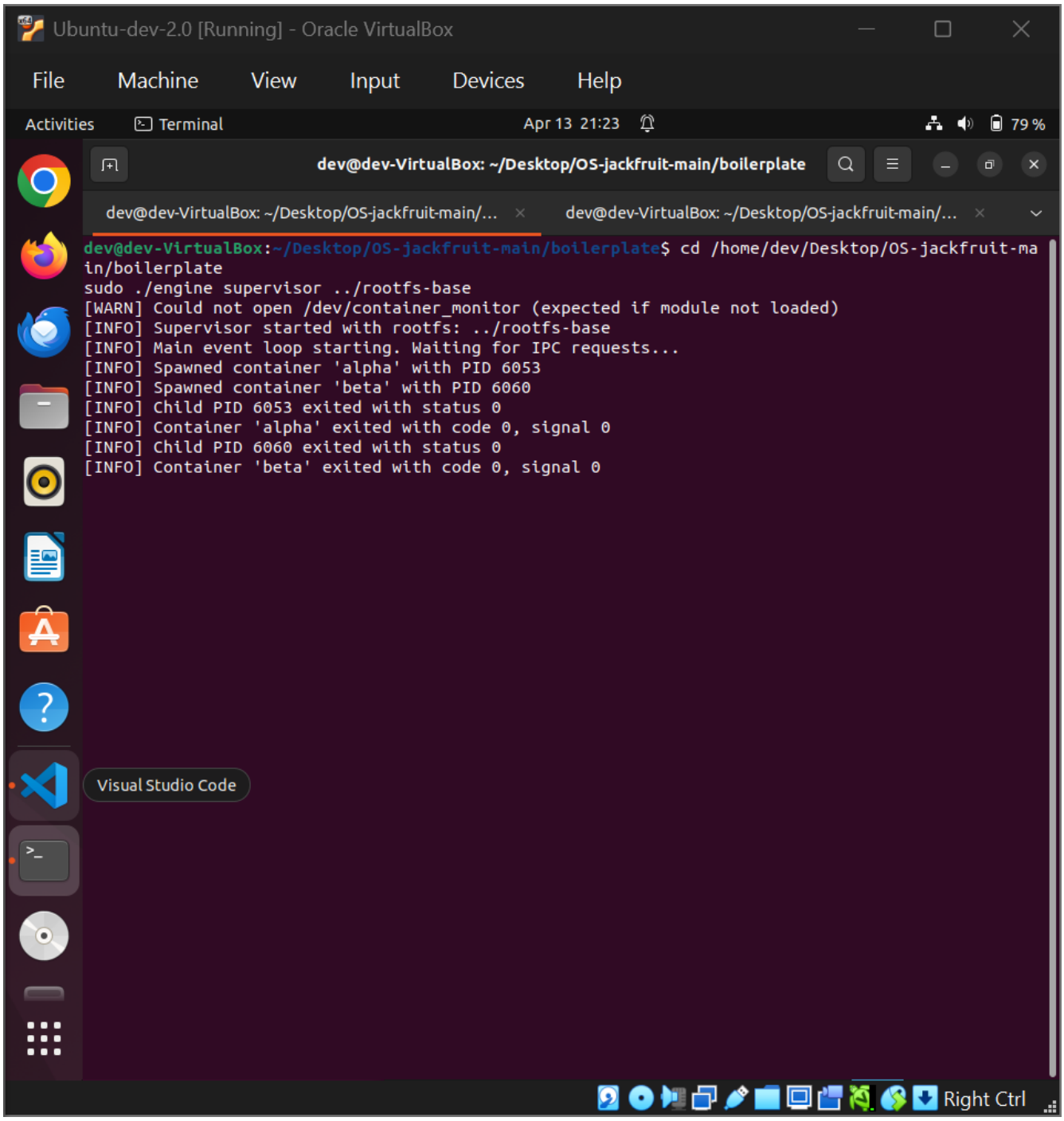
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/... | dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main

```
  2. Spawn 2 containers (alpha and beta)
  3. Show container metadata via 'ps' command
  4. Let containers run for ~10 seconds
  5. Gracefully shut down all containers

Building engine...
make: Nothing to be done for 'all'.

Starting supervisor with rootfs-base...
Commands will be:
  - start alpha ./rootfs-alpha 'sleep 20'
  - start beta ./rootfs-beta 'sleep 20'
  - ps (to view all containers)
  - quit (to shut down)

=== SUPERVISOR OUTPUT BEGINS ===

[WARN] Could not open /dev/container_monitor (expected if module not loaded)
[INFO] Supervisor started with rootfs: ../rootfs-base
[INFO] Main event loop starting. Enter container commands (format: start id rootfs cmd) or 'quit
' to exit.

supervisor> [INFO] Spawned container 'alpha' with PID 5966

supervisor> [INFO] Spawned container 'beta' with PID 5971

supervisor> Container Metadata:
Visual Studio Code      PID        State          Exit
beta                    5971       running        0
alpha                   5966       running        0

supervisor> [INFO] Supervisor shutting down...
[INFO] Terminating container 'beta' (PID 5971)
[INFO] Terminating container 'alpha' (PID 5966)
[INFO] Supervisor exited cleanly

=== TEST COMPLETE ===
Check logs/ directory for per-container logs if implemented
total 8
drwxr-xr-x 2 root root 4096 Apr 13 20:36 .
drwx------ 3 dev  dev  4096 Apr 13 21:05 ..
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main$
```

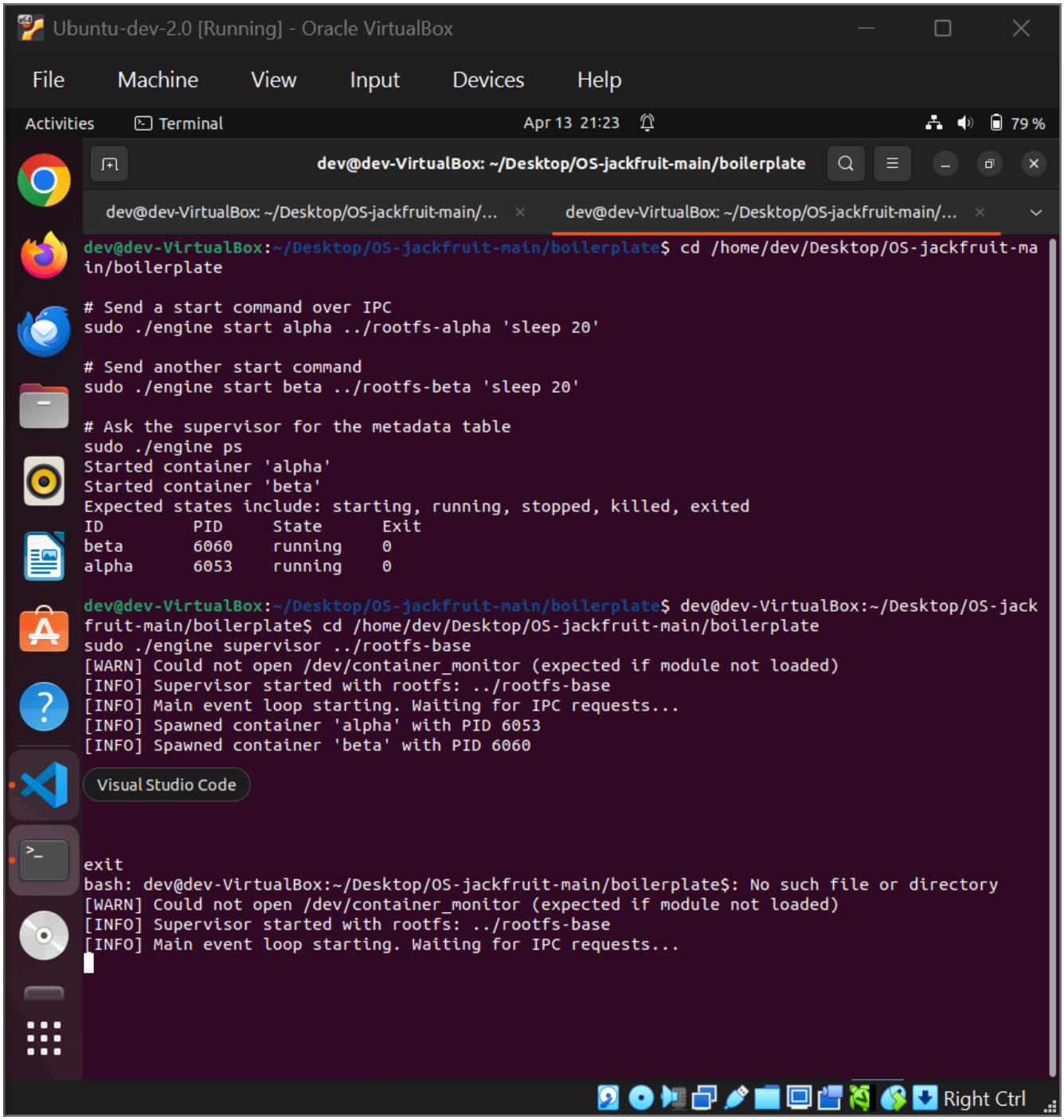
Right Ctrl

Phase 3: The CLI & Control IPC (Task 2)

Terminal1:

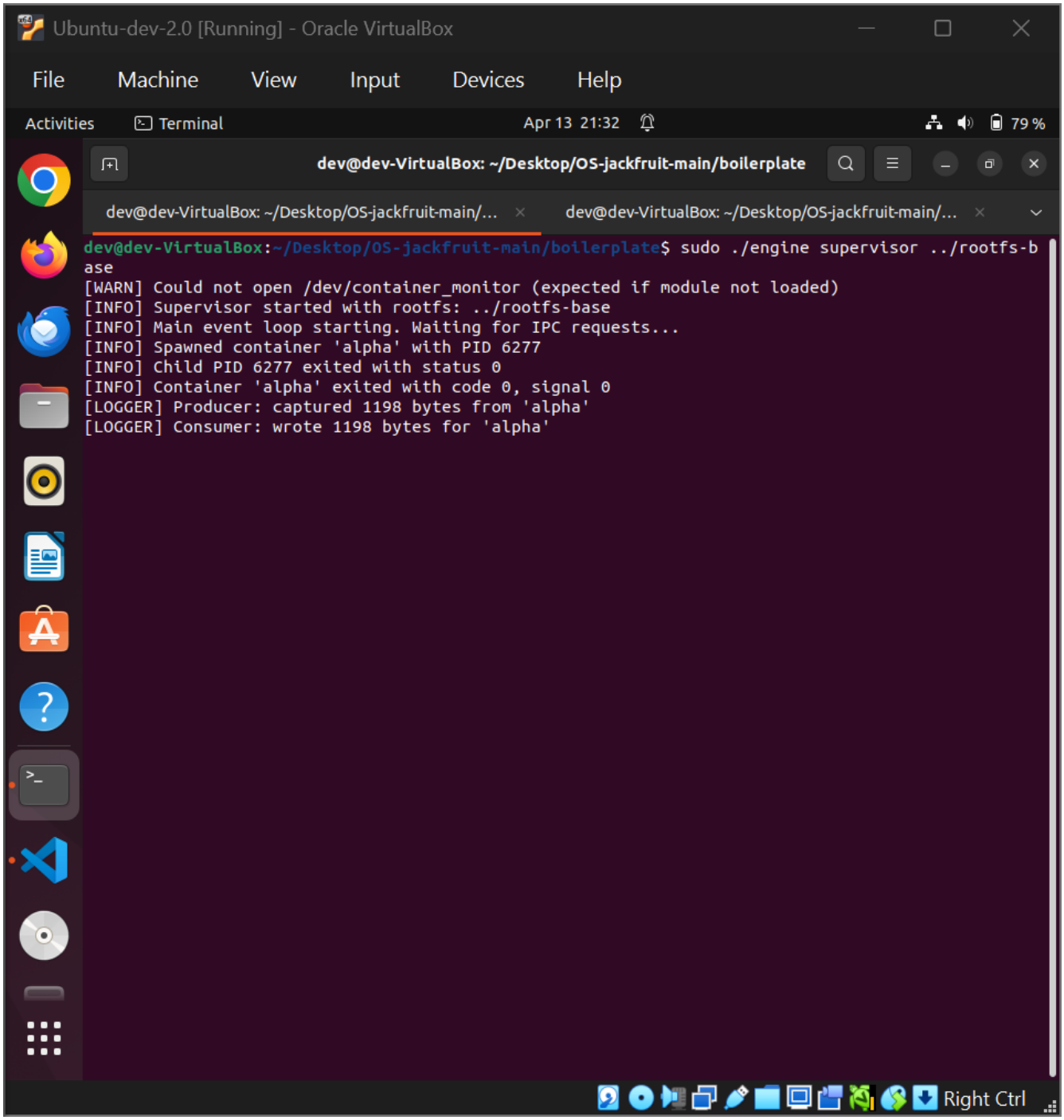


TERMINAL-2

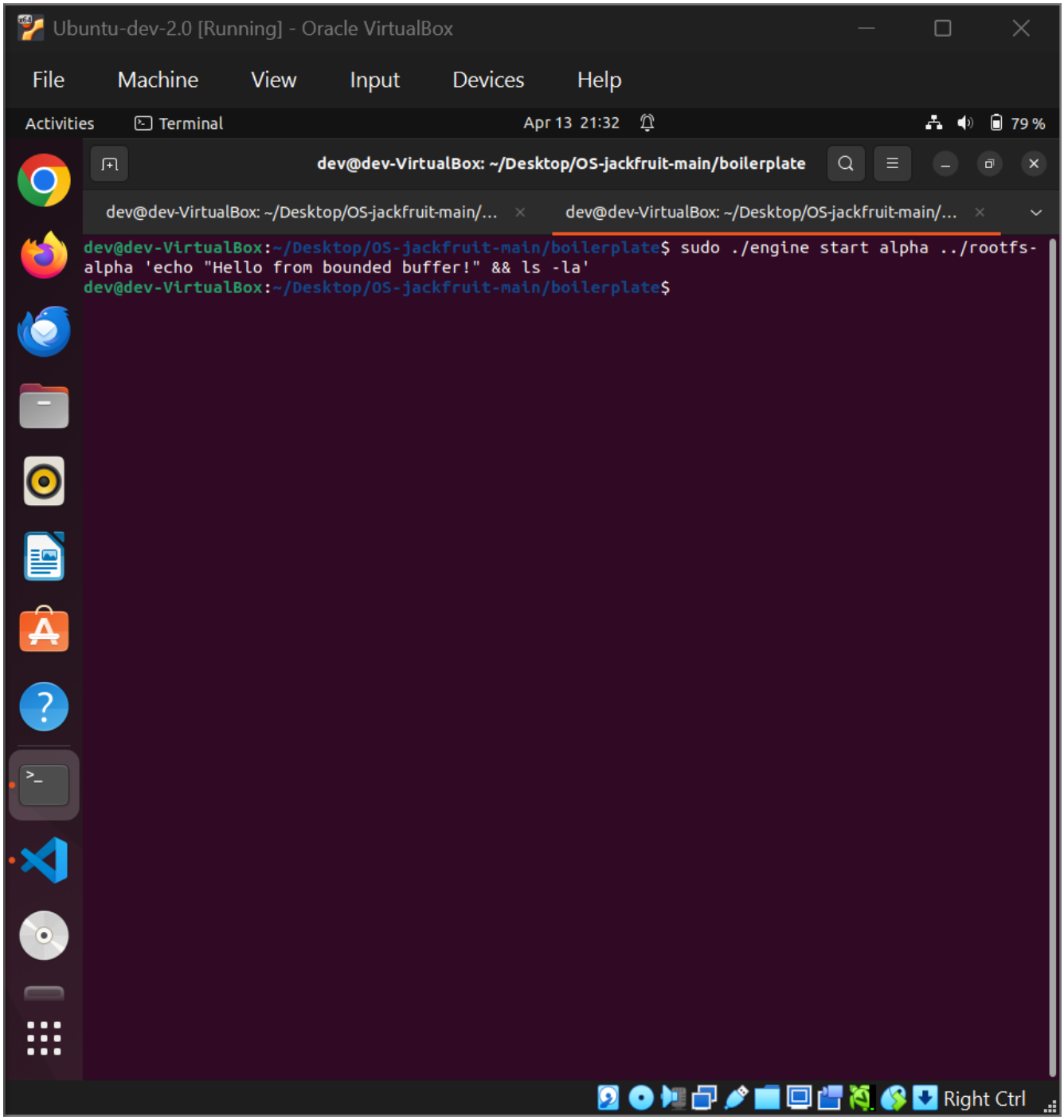

Ubuntu-dev-2.0 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Apr 13 21:23 79 %
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/boilerplate
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/...
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/...
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ cd /home/dev/Desktop/OS-jackfruit-main/boilerplate
# Send a start command over IPC
sudo ./engine start alpha ../rootfs-alpha 'sleep 20'
# Send another start command
sudo ./engine start beta ../rootfs-beta 'sleep 20'
# Ask the supervisor for the metadata table
sudo ./engine ps
Started container 'alpha'
Started container 'beta'
Expected states include: starting, running, stopped, killed, exited
ID PID State Exit
beta 6060 running 0
alpha 6053 running 0
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ cd /home/dev/Desktop/OS-jackfruit-main/boilerplate
sudo ./engine supervisor ../rootfs-base
[WARN] Could not open /dev/container_monitor (expected if module not loaded)
[INFO] Supervisor started with rootfs: ../rootfs-base
[INFO] Main event loop starting. Waiting for IPC requests...
[INFO] Spawned container 'alpha' with PID 6053
[INFO] Spawned container 'beta' with PID 6060
Visual Studio Code
exit
bash: dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$: No such file or directory
[WARN] Could not open /dev/container_monitor (expected if module not loaded)
[INFO] Supervisor started with rootfs: ../rootfs-base
[INFO] Main event loop starting. Waiting for IPC requests...
Right Ctrl

Phase 4: Bounded-Buffer Logging (Task 3)

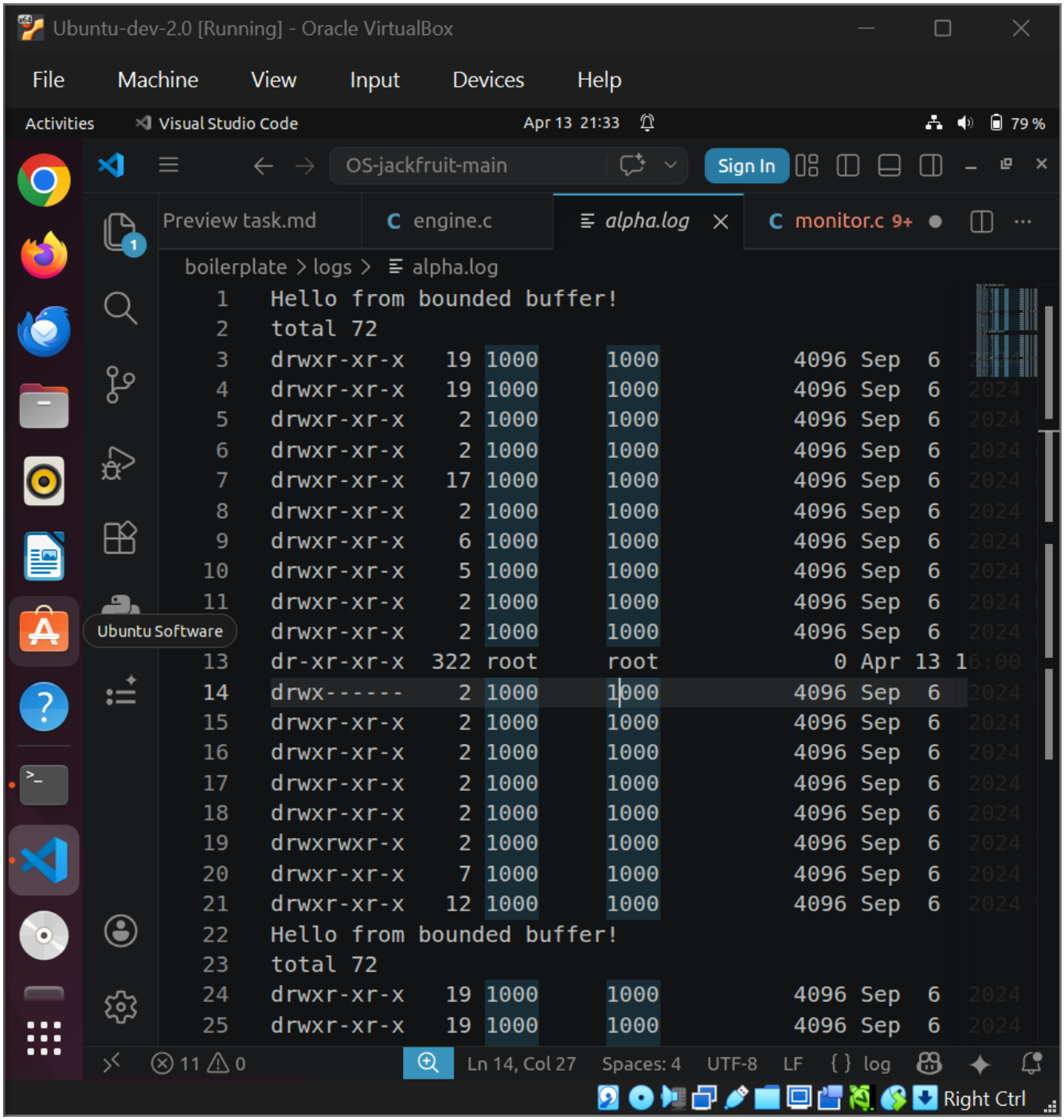
TERMINAL-1:



TERMINAL-2:

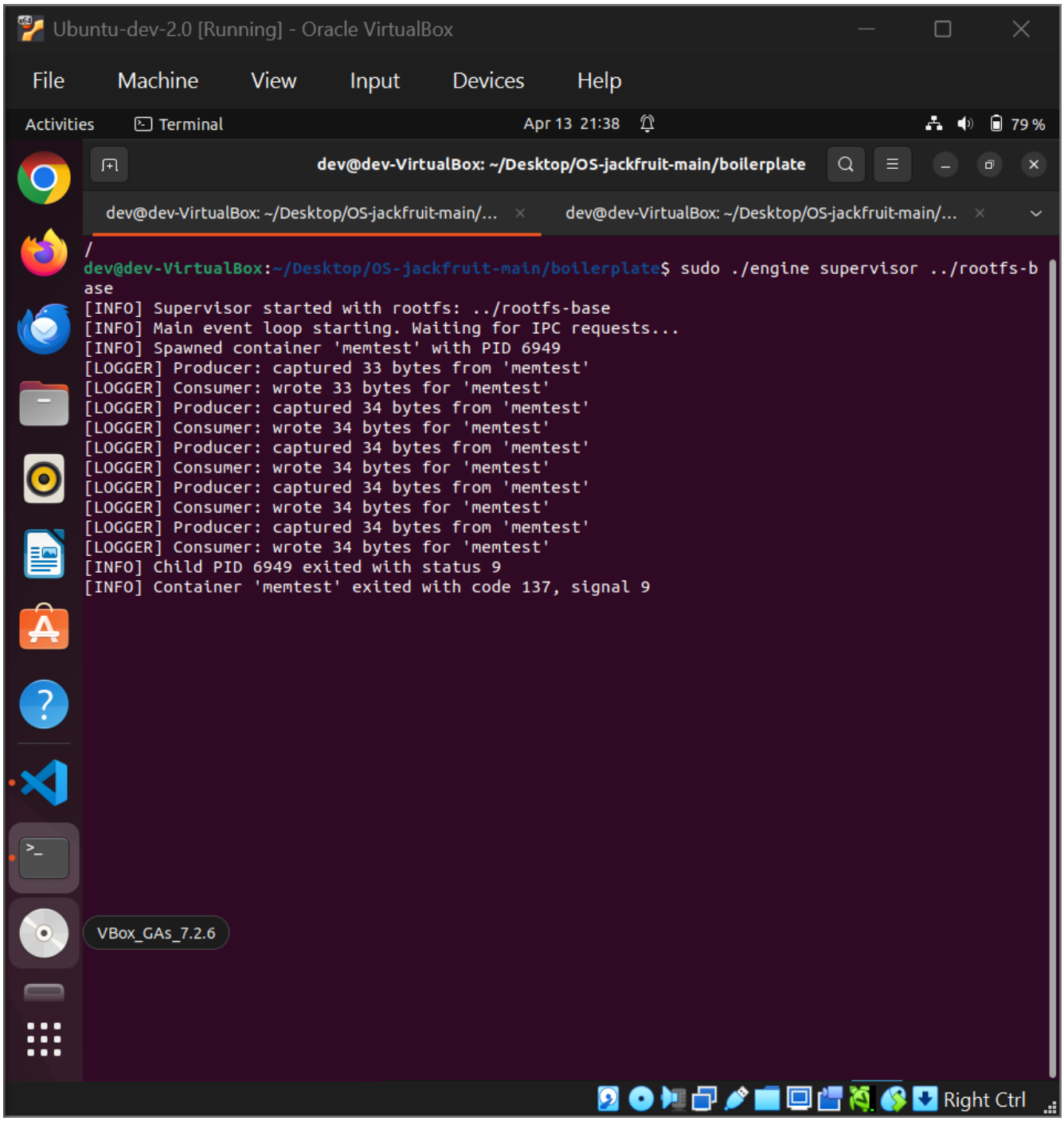

Ubuntu-dev-2.0 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Activities
Terminal
Apr 13 21:32
79 %
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/boilerplate
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/...
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/...
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ sudo ./engine start alpha ../rootfs-alpha 'echo "Hello from bounded buffer!" && ls -la'
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$
Right Ctrl

FROM THE LOGS:

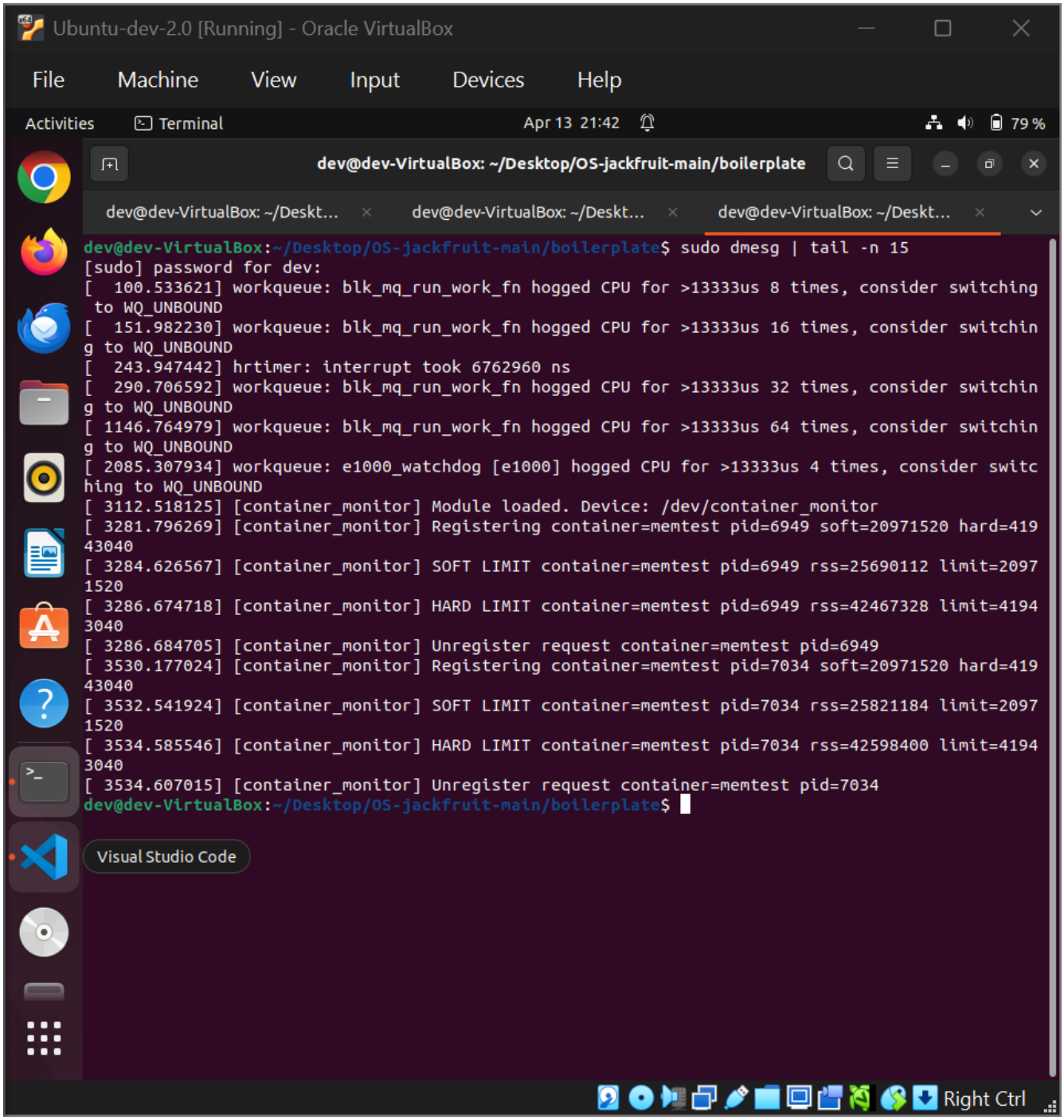


Phase 5: Kernel Memory Monitor (`monitor.c`)

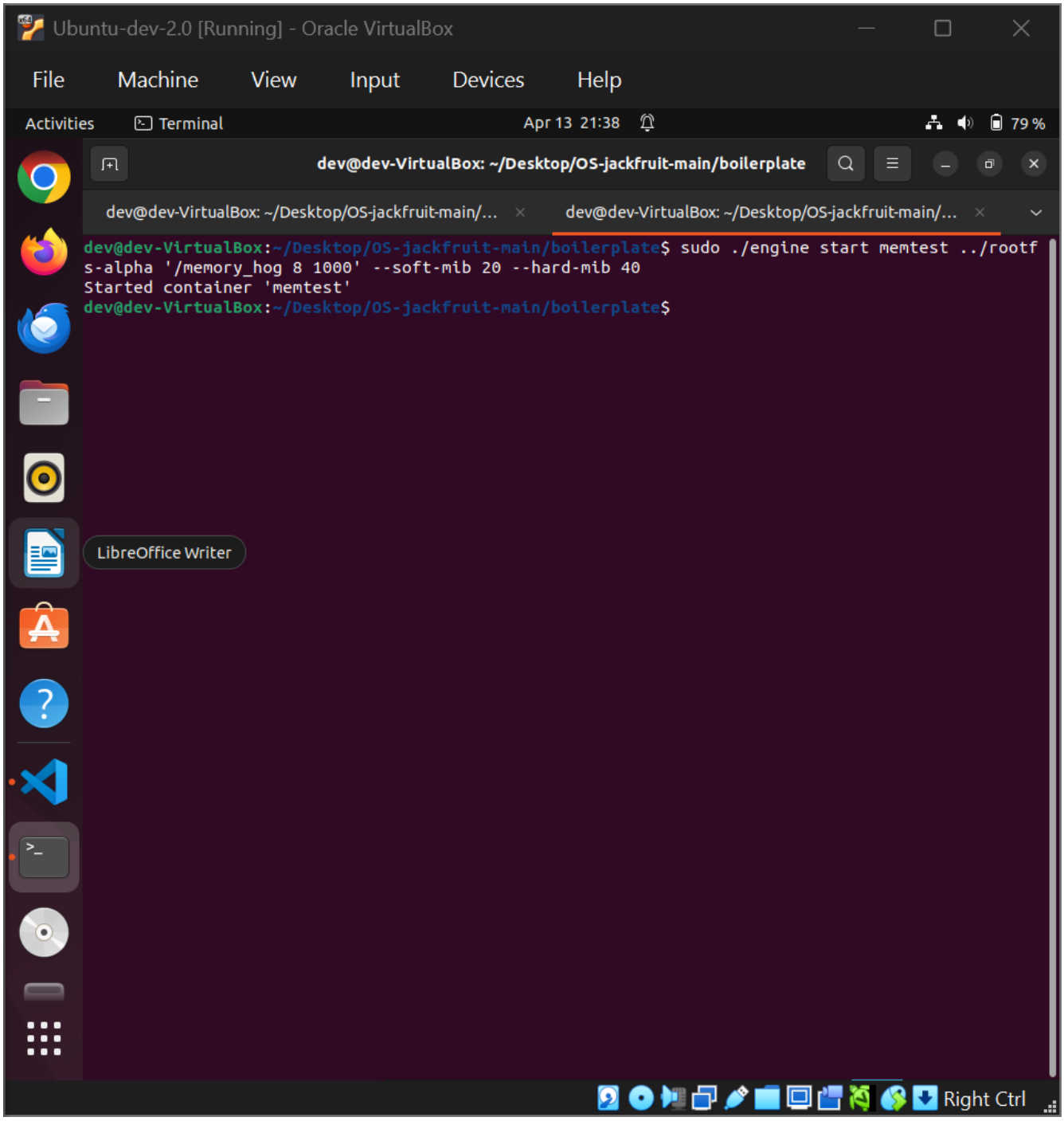
TERMINAL-1:



TERMINAL-3:

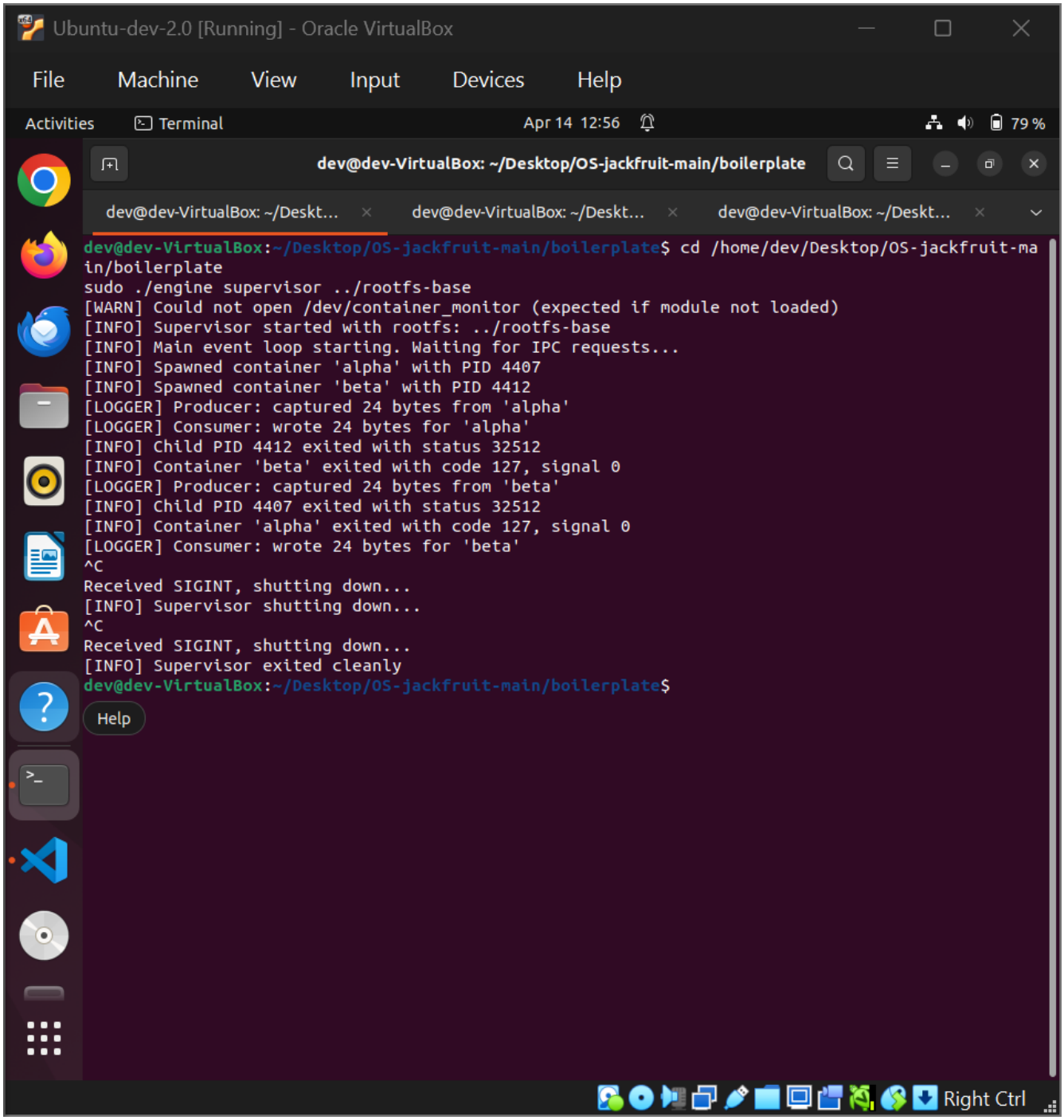


TERMINAL-2:

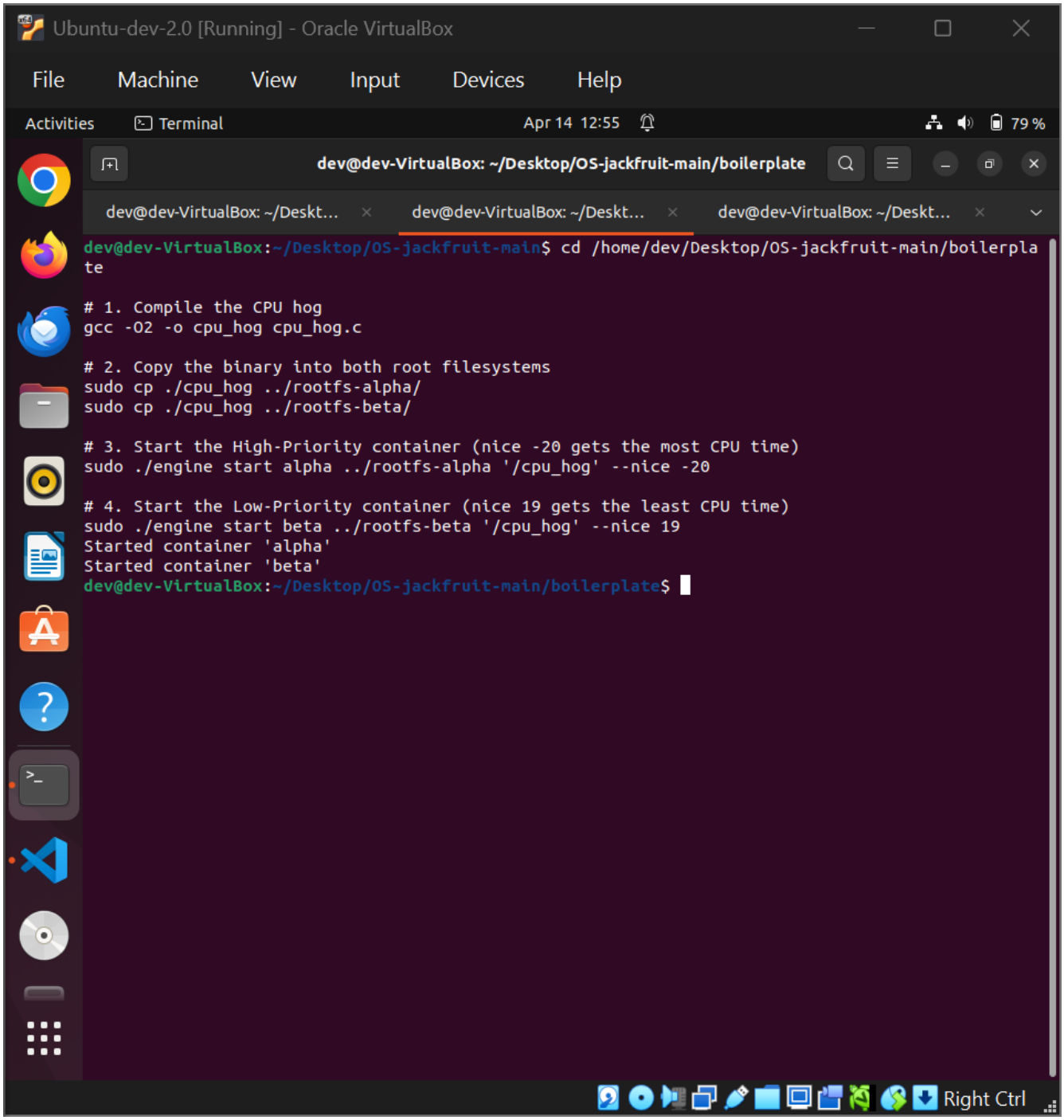

Ubuntu-dev-2.0 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Activities
Terminal
Apr 13 21:38
79 %
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/boilerplate
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/...
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/...
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ sudo ./engine start memtest ../rootf
s-alpha '/memory_hog 8 1000' --soft-mib 20 --hard-mib 40
Started container 'memtest'
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$
LibreOffice Writer
Right Ctrl

Phase 6: Scheduler Experiments (Task 5)

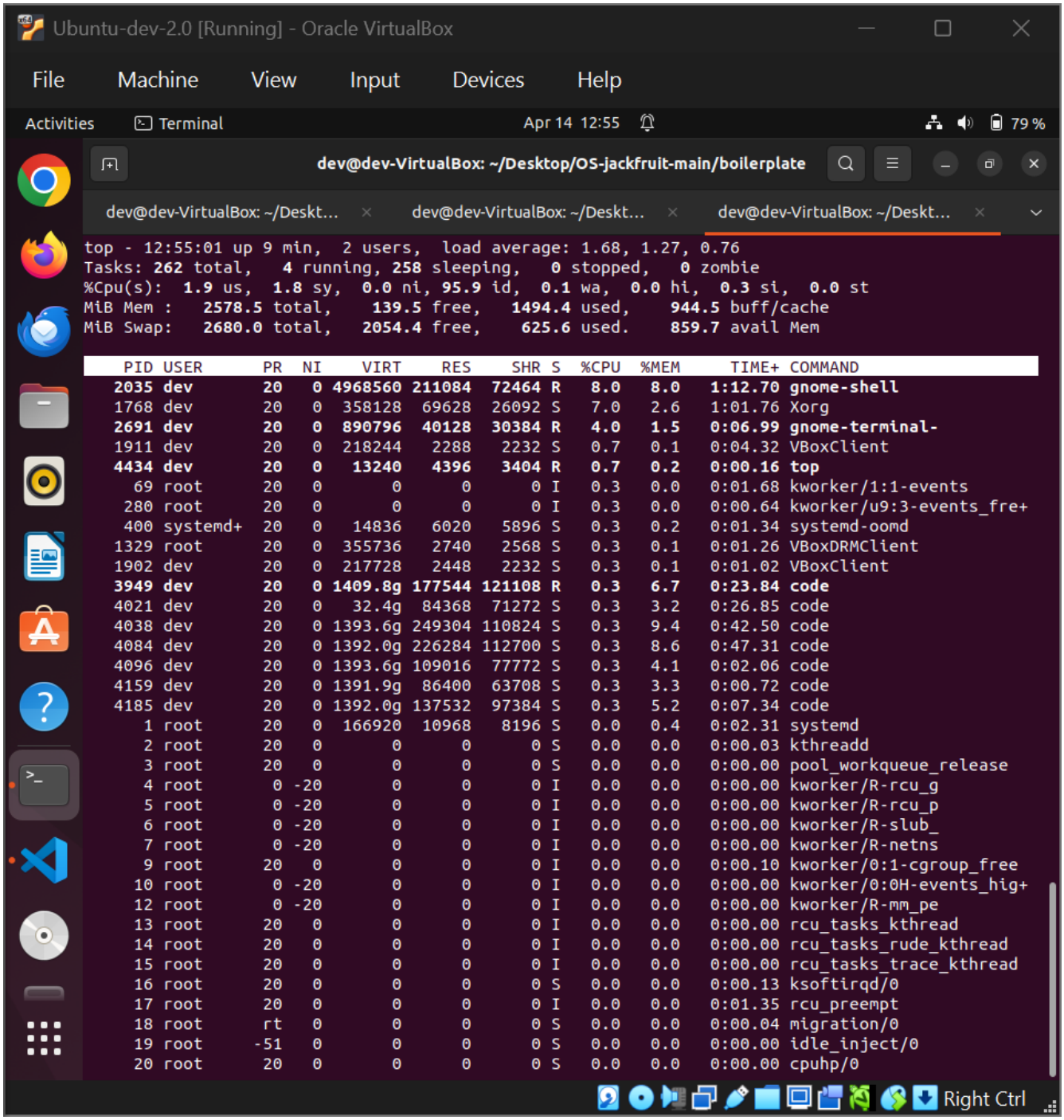
TERMINAL-1:



TERMINAL-2:

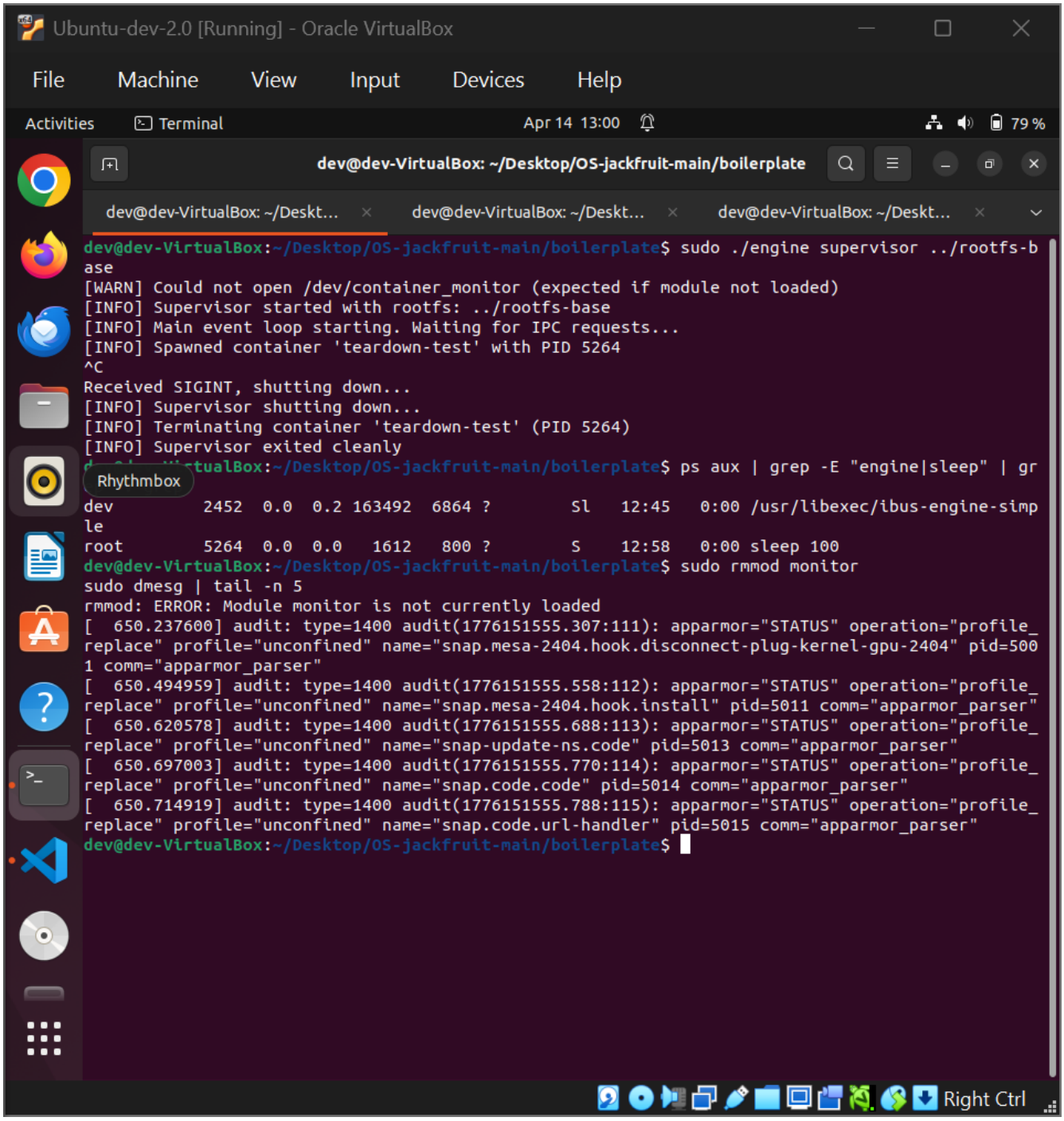

Ubuntu-dev-2.0 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Apr 14 12:55 79 %
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/boilerplate
dev@dev-VirtualBox: ~/Deskt...
dev@dev-VirtualBox: ~/Deskt...
dev@dev-VirtualBox: ~/Deskt...
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main$ cd /home/dev/Desktop/OS-jackfruit-main/boilerplate
1. Compile the CPU hog
gcc -O2 -o cpu_hog cpu_hog.c
# 2. Copy the binary into both root filesystems
sudo cp ./cpu_hog ../rootfs-alpha/
sudo cp ./cpu_hog ../rootfs-beta/
# 3. Start the High-Priority container (nice -20 gets the most CPU time)
sudo ./engine start alpha ../rootfs-alpha '/cpu_hog' --nice -20
# 4. Start the Low-Priority container (nice 19 gets the least CPU time)
sudo ./engine start beta ../rootfs-beta '/cpu_hog' --nice 19
Started container 'alpha'
Started container 'beta'
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$
Right Ctrl

TERMINAL-3:

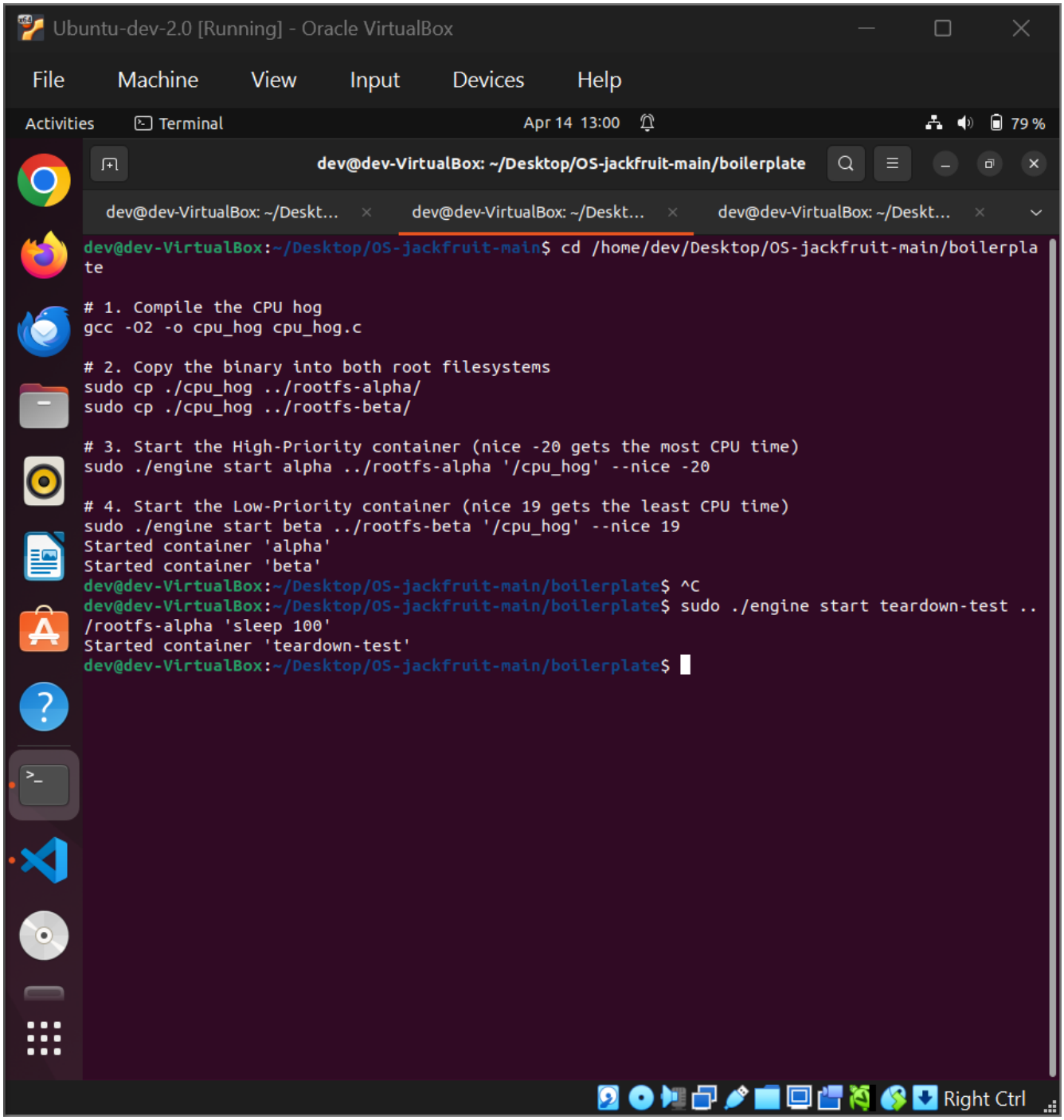

Ubuntu-dev-2.0 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Apr 14 12:55 79 %
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/boilerplate
dev@dev-VirtualBox: ~/Deskt...
dev@dev-VirtualBox: ~/Deskt...
dev@dev-VirtualBox: ~/Deskt...
top - 12:55:01 up 9 min, 2 users, load average: 1.68, 1.27, 0.76
Tasks: 262 total, 4 running, 258 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.9 us, 1.8 sy, 0.0 ni, 95.9 id, 0.1 wa, 0.0 hi, 0.3 si, 0.0 st
MiB Mem : 2578.5 total, 139.5 free, 1494.4 used, 944.5 buff/cache
MiB Swap: 2680.0 total, 2054.4 free, 625.6 used. 859.7 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
2035 dev 20 0 4968560 211084 72464 R 8.0 8.0 1:12.70 gnome-shell
1768 dev 20 0 358128 69628 26092 S 7.0 2.6 1:01.76 Xorg
2691 dev 20 0 890796 40128 30384 R 4.0 1.5 0:06.99 gnome-terminal-
1911 dev 20 0 218244 2288 2232 S 0.7 0.1 0:04.32 VBoxClient
4434 dev 20 0 13240 4396 3404 R 0.7 0.2 0:00.16 top
69 root 20 0 0 0 0 I 0.3 0.0 0:01.68 kworker/1:1-events
280 root 20 0 0 0 0 I 0.3 0.0 0:00.64 kworker/u9:3-events_fre+
400 systemd+ 20 0 14836 6020 5896 S 0.3 0.2 0:01.34 systemd-oomd
1329 root 20 0 355736 2740 2568 S 0.3 0.1 0:01.26 VBoxDRMClient
1902 dev 20 0 217728 2448 2232 S 0.3 0.1 0:01.02 VBoxClient
3949 dev 20 0 1409.8g 177544 121108 R 0.3 6.7 0:23.84 code
4021 dev 20 0 32.4g 84368 71272 S 0.3 3.2 0:26.85 code
4038 dev 20 0 1393.6g 249304 110824 S 0.3 9.4 0:42.50 code
4084 dev 20 0 1392.0g 226284 112700 S 0.3 8.6 0:47.31 code
4096 dev 20 0 1393.6g 109016 77772 S 0.3 4.1 0:02.06 code
4159 dev 20 0 1391.9g 86400 63708 S 0.3 3.3 0:00.72 code
4185 dev 20 0 1392.0g 137532 97384 S 0.3 5.2 0:07.34 code
1 root 20 0 166920 10968 8196 S 0.0 0.4 0:02.31 systemd
2 root 20 0 0 0 0 S 0.0 0.0 0:00.03 kthreadd
3 root 20 0 0 0 0 S 0.0 0.0 0:00.00 pool_workqueue_release
4 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-rcu_g
5 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-rcu_p
6 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-slub_
7 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-netns
9 root 20 0 0 0 0 I 0.0 0.0 0:00.10 kworker/0:1-cgroup_free
10 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/0:0H-events_hig+
12 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-mm_pe
13 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_kthread
14 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_rude_kthread
15 root 20 0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_trace_kthread
16 root 20 0 0 0 0 S 0.0 0.0 0:00.13 ksoftirqd/0
17 root 20 0 0 0 0 I 0.0 0.0 0:01.35 rcu_preempt
18 root rt 0 0 0 0 S 0.0 0.0 0:00.04 migration/0
19 root -51 0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/0
20 root 20 0 0 0 0 S 0.0 0.0 0:00.00 cpuhp/0
Right Ctrl

Phase 7: Clean Teardown (Task 6)

TERMINAL-1:

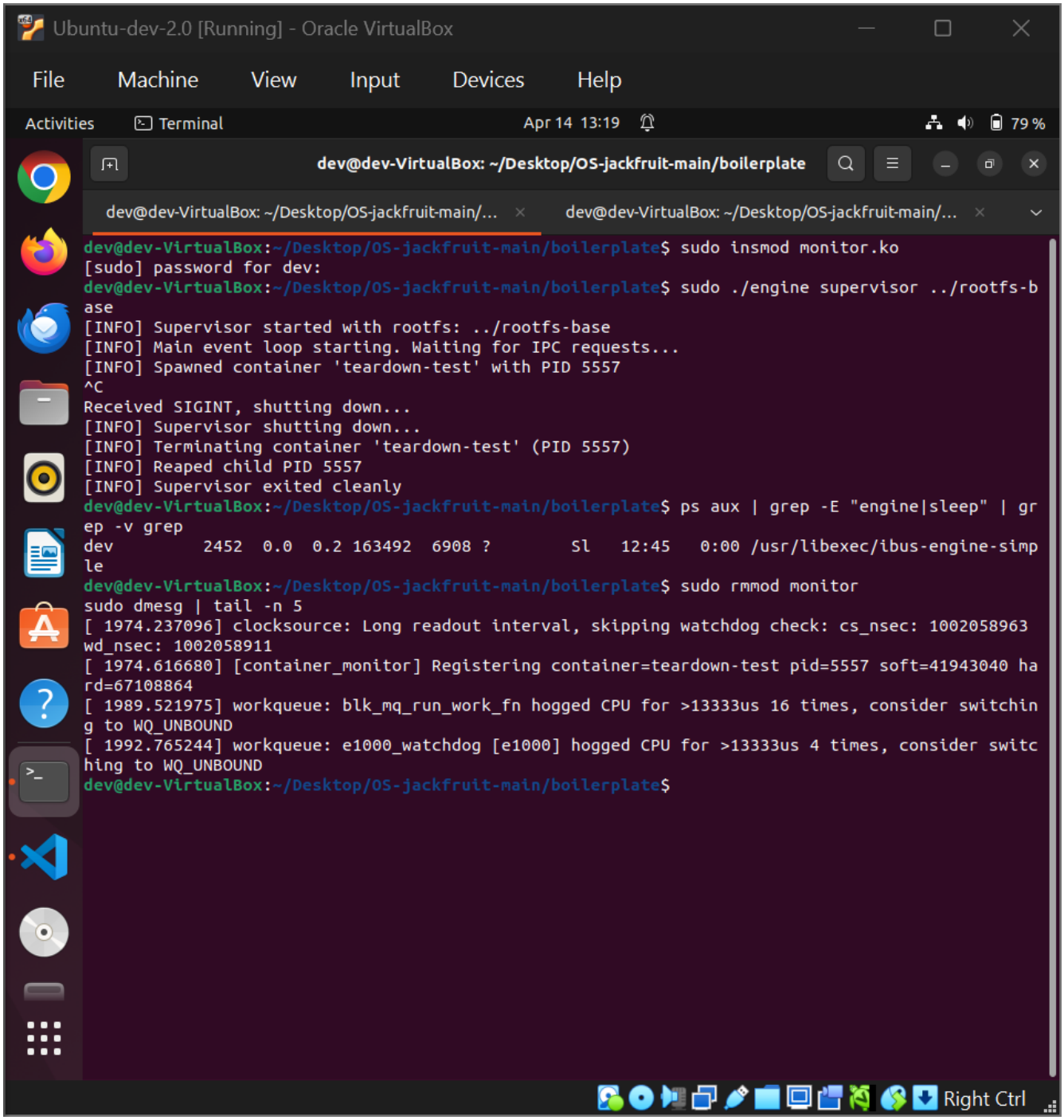


TERMINAL-2:


Ubuntu-dev-2.0 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities Terminal Apr 14 13:00 79 %
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/boilerplate
dev@dev-VirtualBox: ~/Deskt...
dev@dev-VirtualBox: ~/Deskt...
dev@dev-VirtualBox: ~/Deskt...
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main$ cd /home/dev/Desktop/OS-jackfruit-main/boilerplate
1. Compile the CPU hog
gcc -O2 -o cpu_hog cpu_hog.c
# 2. Copy the binary into both root filesystems
sudo cp ./cpu_hog ../rootfs-alpha/
sudo cp ./cpu_hog ../rootfs-beta/
# 3. Start the High-Priority container (nice -20 gets the most CPU time)
sudo ./engine start alpha ../rootfs-alpha '/cpu_hog' --nice -20
# 4. Start the Low-Priority container (nice 19 gets the least CPU time)
sudo ./engine start beta ../rootfs-beta '/cpu_hog' --nice 19
Started container 'alpha'
Started container 'beta'
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ ^C
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ sudo ./engine start teardown-test ../rootfs-alpha 'sleep 100'
Started container 'teardown-test'
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$
Right Ctrl

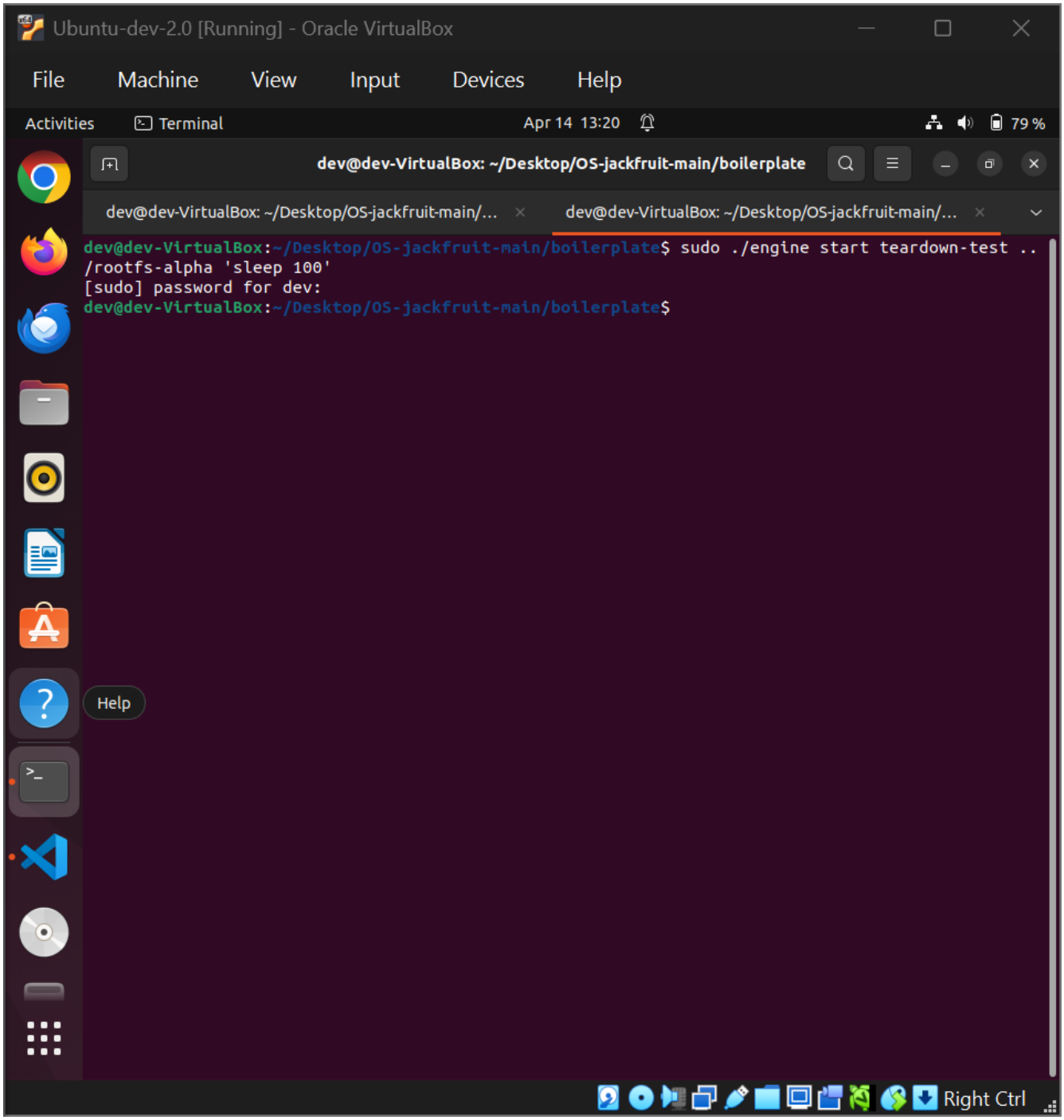
Phase 8: Final Submission Wrap-up

TERMINAL-1:



```
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ sudo insmod monitor.ko
[sudo] password for dev:
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ sudo ./engine supervisor ../rootfs-b
ase
[INFO] Supervisor started with rootfs: ../rootfs-base
[INFO] Main event loop starting. Waiting for IPC requests...
[INFO] Spawned container 'teardown-test' with PID 5557
^C
Received SIGINT, shutting down...
[INFO] Supervisor shutting down...
[INFO] Terminating container 'teardown-test' (PID 5557)
[INFO] Reaped child PID 5557
[INFO] Supervisor exited cleanly
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ ps aux | grep -E "engine|sleep" | gr
ep -v grep
dev         2452  0.0  0.2 163492  6908 ?        Sl   12:45   0:00 /usr/libexec/ibus-engine-simp
le
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ sudo rmmod monitor
sudo dmesg | tail -n 5
[ 1974.237096] clocksource: Long readout interval, skipping watchdog check: cs_nsec: 1002058963
wd_nsec: 1002058911
[ 1974.616680] [container_monitor] Registering container=teardown-test pid=5557 soft=41943040 ha
rd=67108864
[ 1989.521975] workqueue: blk_mq_run_work_fn hogged CPU for >13333us 16 times, consider switchin
g to WQ_UNBOUND
[ 1992.765244] workqueue: e1000_watchdog [e1000] hogged CPU for >13333us 4 times, consider switc
hing to WQ_UNBOUND
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$
```

TERMINAL-2:


Ubuntu-dev-2.0 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Activities
Terminal
Apr 14 13:20
79 %
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/boilerplate
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/...
dev@dev-VirtualBox: ~/Desktop/OS-jackfruit-main/...
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$ sudo ./engine start teardown-test ../rootfs-alpha 'sleep 100'
[sudo] password for dev:
dev@dev-VirtualBox:~/Desktop/OS-jackfruit-main/boilerplate$
Help
Right Ctrl

LIST OF FIGURES (Output Screenshots)

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